MATH 301: Homework 5

Aren Vista

April 21, 2026

Theorem 4.2.4: Algebraic Limit Theorem
Let $A \subseteq \mathbb{R}$, let f and g be functions on A to $\mathbb{R}$, let $c \in \mathbb{R}$ be a cluster point of A. Let $b \in \mathbb{R}$.

(a) *If $\lim_{x\to c} f = L$ and $\lim_{x\to c} g = M$, then:*

$$\lim_{x\to c}(f+g) = L + M, \qquad \lim_{x\to c}(f-g) = L - M,$$
$$\lim_{x\to c}(fg) = LM, \qquad \lim_{x\to c}(bf) = bL.$$

(b) *If $h : A \to \mathbb{R}$, if $h(x) \neq 0$ for all $x \in A$, and if $\lim_{x\to c} h = H \neq 0$, then*

$$\lim_{x\to c}\left(\frac{f}{h}\right) = \frac{L}{H}.$$

1 Section 4.2 - 1c

Apply Theorem 4.2.4 to determine the following limit:

$$\lim_{x\to 2}\left(\frac{1}{x+1} - \frac{1}{2x}\right) \quad (x > 0)$$

2 Section 4.2 - 2b

Determine the following limits and state which theorems are used in each case

$$\lim_{x\to 2}\frac{x^2-4}{x-2} \quad (x > 0)$$

3 Section 4.2 - 2d

Determine the following limits and state which theorems are used in each case

$$\lim_{x\to 1}\frac{\sqrt{x}-1}{x-1} \quad (x > 0)$$

4 Section 4.2 - 4

Prove that $\lim_{x\to 0} cos(\frac{1}{x})$ does not exist but $\lim_{x\to 0} xcos(\frac{1}{x}) = 0$.

5 Section 4.2 - 10

Give examples of functions f and g such that f and g do not have limits at a point c, but such that both $f + g$ and fg have limits at c.

6 Section 5.1 - 4a

7 Section 5.1 - 4b

8 Section 5.1 - 5

9 Section 5.1 - 6

10 Section 5.1 - 10

11 Section 5.1 - 14

12 Section 5.2 - 2

13 Section 5.2 - 3

14 Section 5.2 - 5

15 Section 5.2 - 7

16 Section 5.2 - 8

17 Section 5.2 - 12